8. Monoalphabetic Cipher using keyword

#include <stdio.h> #include <string.h> #include <ctype.h>

int main() { char keyword[100], plaintext[500], cipher[27]; char alphabet[] = "abcdefghijklmnopqrstuvwxyz"; int used[26] = {0}; int i, j = 0, len;

printf("Enter keyword: ");
scanf("%s", keyword);

/* Generate cipher sequence from keyword */
len = strlen(keyword);

for (i = 0; i < len; i++)
{
   char ch = tolower(keyword[i]);

   if (ch >= 'a' && ch <= 'z' && used[ch - 'a'] == 0)
   {
      cipher[j++] = ch;
      used[ch - 'a'] = 1;
   }
}

/* Add unused letters in normal order */
for (i = 0; i < 26; i++)
{
   if (used[i] == 0)
   {
      cipher[j++] = alphabet[i];
   }
}

cipher[26] = '\0';

printf("\nPlain : ");
for (i = 0; i < 26; i++)
   printf("%c ", alphabet[i]);

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printf("\nCipher: ");
for (i = 0; i < 26; i++)
   printf("%c ", cipher[i]);

/* Read plaintext */
printf("\n\nEnter plaintext: ");
getchar();
fgets(plaintext, sizeof(plaintext), stdin);

/* Encrypt */
printf("Ciphertext: ");

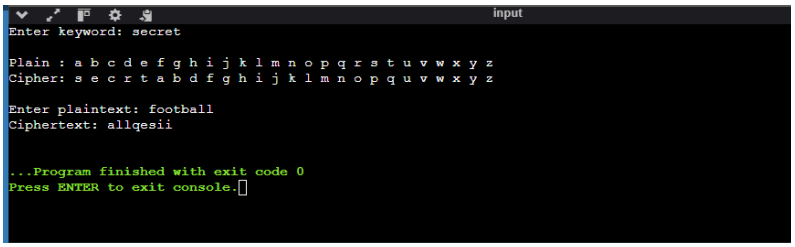
for (i = 0; plaintext[i] != '\0'; i++)
{
   char ch = plaintext[i];

   if (ch >= 'a' && ch <= 'z')
   {
      printf("%c", cipher[ch - 'a']);
   }
   else if (ch >= 'A' && ch <= 'Z')
   {
      printf("%c", toupper(cipher[ch - 'A']));
   }
   else
   {
      printf("%c", ch);
   }
}

return 0;

}
```

Output


input
Enter keyword: secret
Plain : a b c d e f g h i j k l m n o p q r s t u v w x y z
Cipher: s e c r t a b d f g h i j k l m n o p q u v w x y z
Enter plaintext: football
Ciphertext: allqesii
...Program finished with exit code 0
Press ENTER to exit console.